

LOOKUP FUNCTIONS

Lookup functions typically work by finding (or matching) a *lookup value* in a column and returning a related value in another column from the same row:

This is known as a **fact** (or data) table

Flight	Airline	Destination	Destination City
AA258	American Airlines	MIA	
US726	US Airways	PIT	
VX412	Virgin America	JFK	
AA28	American Airlines	MIA	
OO6528	Skywest	PHX	
AS461	Alaska Airlines	SEA	
AA219	American Airlines	SFO	
AA10	American Airlines	JFK	
VX406	Virgin America	JFK	
VX927	Virgin America	SFO	
AA1064	American Airlines	MIA	
US1927	US Airways	PHL	
AA118	American Airlines	JFK	
B6688	JetBlue	BOS	
B6288	JetBlue	BOS	

This is known as a **dimension** (or lookup) table

IATA Code	Airport Name	City
BOS	Logan International Airport	Boston
JFK	John F. Kennedy International Airport	New York
MIA	Miami International Airport	Miami
PHL	Philadelphia International Airport	Philadelphia
PHX	Phoenix Sky Harbor International Airport	Phoenix
PIT	Pittsburgh International Airport	Pittsburgh
SEA	Seattle-Tacoma International Airport	Seattle
SFO	San Francisco International Airport	San Francisco

lookup value

This is a **primary key** (no duplicates)

This is a **foreign key**

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related value

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THE XLOOKUP FUNCTION

XLOOKUP()

Returns a value from a result range in the position where a match was found in a lookup range

=XLOOKUP(lookup_value, lookup_array, return_array, [if_not_found], [match_mode] , [search_mode])

Which value are you looking to match?

Where are you trying to find the match for it?

Where are the values
you want to return?

What if the lookup value isn't found?

Do you want an exact or approximate match?

*Do you want to search
top-down or bottom-up?*

D2					=XLOOKUP(C2,\$F\$2:\$F\$9,\$H\$2:\$H\$9)			
	A	B	C	D	E	F	G	H
1	Flight	Airline	Destination	Destination City		IATA Code	Airport Name	City
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4	VX412	Virgin America	JFK	New York		MIA	Miami International Airport	Miami
5	AA28	American Airlines	MIA	Miami		PHL	Philadelphia International Airport	Philadelphia
6	OO6528	Skywest	PHX	Phoenix		PHX	Phoenix Sky Harbor International Airport	Phoenix
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8	AA219	American Airlines	SFO	San Francisco		SEA	Seattle-Tacoma International Airport	Seattle
9	AA10	American Airlines	JFK	New York		SFO	San Francisco International Airport	San Francisco
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